KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering
Spring 2025 Final Examination
Program: B.Sc. in CSE (Batch: 17^{th}) 2^{nd} Year 1^{st} Semester
Course Title: Linear Algebra and Matrix Analysis
Course Code: CSE 0541-2101

PART-A: MCQ

Time: 10 Minutes | Total Marks: 10

Student's ID:

Date: | Invigilator's Signature:

(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

CLO 3	Formulate real problems in mathematical way regarding vector spaces, Complex function and Number System, Consistent, inconsistent, eigenvalues and eigenvectors and Fourier series	07

Cognitive Analyzing

1. Eigen values of a real matrix are always ______number.
   - A) real
   - B) positive
   - C) real and imaginary
   - D) negative
   - E) imaginary

Cognitive Evaluating

2. Which is correct eigen vectors for the matrix $\begin{bmatrix} 0 & 0 & a \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$?
   - A) $(0, a, 0)$
   - B) $(a, a, 0)$
   - C) $(0, a, a)$
   - D) $(a, 0, 0)$
   - E) $(0, 0, a)$

Cognitive Understanding

3. A set of vectors in a vector space is called____.
   - A) coordinate
   - B) Basis
   - C) Space
   - D) Unit
   - E) Dimension

Cognitive Remembering

4. What is the value of the dimension of subspace, spanned by $\begin{pmatrix} 1 & -5 \\ -4 & 2 \end{pmatrix}$, $\begin{pmatrix} 1 & 1 \\ -1 & 5 \end{pmatrix}$ and $\begin{pmatrix} 2 & -4 \\ -5 & 7 \end{pmatrix}$?
   - A) 1
   - B) 2
   - C) 3
   - D) 4
   - E) 6

Cognitive Understanding

5. Which solution is required for a system of equation is said to be inconsistent?

A) one
B) two
C) finite
D) no
E) infinite

Cognitive Understanding

6. If a matrix A has rank 3 and is a 4×5 matrix, what is the nullity of A?

A) One
B) Two
C) Three
D) Four
E) Five

Cognitive Remembering

7. The system of linear equations $4x + 2y = 7$, $2x + y = 6$ has ________ solutions.

A) unique
B) exactly two distinct
C) zero
D) no
E) infinite

CLO4	apply the reduced (row) Echelon form of a matrix for determine linear combination, linearly dependent and independent, basis and dimension.	02

Cognitive Understanding

8. Let u, v, w be three non-zero vectors which are linearly independent, then which is the right statement?

A) u is linear combination of v and w
B) B) v is linear combination of u and w
C) C) w is linear combination of u and v
D) D) w is linear combination of u and u
E) E) cannot be determined

Cognitive Applying

9. If $A: R^4 \to R^3$ where, $A = \begin{pmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 5 & -2 \\ 3 & 8 & 13 & -3 \end{pmatrix}$,then what will be the dimension of the kernel of A?

A) 1
B) 2
C) 3
D) 4
E) 5

Cognitive Analyzing

10. If a vector space has two linearly independent vectors, which of the following is true about the dimension of the space?

A) The dimension is at least 2.
B) The dimension is 2 only if the space is finite-dimensional.
C) The dimension is exactly 1.
D) The dimension is infinite.
E) The space is 1-dimensional.

KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering
Summer 2024 Final Examination
Program: B.Sc. in CSE (Batch:16th) 2nd Year 1st Semester
Course Title: Linear Algebra and Matrix Analysis
Course Code: CSE 0541-2101

PART-A: MCQ

Total Marks: 10

Time: 10 Minutes

Student's ID:

Date:

Invigilator's Signature:

(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

CLO2	apply theory and calculation techniques for systems of linear equations (homogeneous/non homogeneous) to determine solvability and provide complete solutions and their structures.	02

Cognitive Analyzing

1. If the system of equations $2x + y = 8$ and $-4x - 2y = -16$ has infinitely many solutions, what can we conclude about these two equations?
   A) Dependent and inconsistent.
   B) Independent and consistent.
   C) Dependent and consistent.
   D) Independent and inconsistent.
   E) They have no relationship to each other.

Cognitive Evaluating

2. Solve the system $2x + y = 6$ and $3x - y = 4$, What is the value of x?
   A) 1
   B) 2
   C) 3
   D) 4
   E) 5

CLO 3	Formulate real problems in mathematical way regarding vector spaces, rank, nullity, Consistent, inconsistent, eigenvalues and eigenvectors and linear transformations.	06

Cognitive Understanding

3. Which of the following statements best describes a subspace?
   A) Subset that includes the zero vector, is closed under addition, and is closed under scalar multiplication.
   B) A set that includes all possible linear combinations of vectors.
   C) A set containing only the zero vector.
   D) Any subset that has at least two linearly independent vectors.
   E) A set that is closed under multiplication but not necessarily under addition.

Cognitive Remembering

4. The rank of a matrix is the number of________.
   A) columns in the matrix.
   B) rows in the matrix.
   C) non-zero rows in its row echelon form.
   D) zeros in the matrix.
   E) columns minus the number of rows.

Cognitive Understanding

5. If a matrix A has rank 3 and is a 4×5 matrix, what is the nullity of A?
   A) One
   B) Two
   C) Three
   D) Four
   E) Five

Cognitive Remembering

6. If a system of equations has three equations and two unknowns, which of the following could make the system consistent?
   A) All equations are identical.
   B) All equations are parallel lines.
   C) All equations intersect at a single point.
   D) The equations have no variables.
   E) Each equation has a different slope.

Cognitive Remembering

7. Which of the following terms refers to a system of linear equations that has no solutions?
   A) Consistent
   B) Independent
   C) Inconsistent
   D) Dependent
   E) Homogeneous

Cognitive Evaluating

8. Given the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, which of the following is an eigenvalue of A?
   A) 0
   B) 1
   C) 2
   D) 3
   E) 4

CLO4	apply the reduced (row) Echelon form of a matrix for determine linear combination, linearly dependent and independent, basis and dimension.	02

Cognitive Applying

9. Which of the following sets of vectors can form a linear combination that produces the vector (6,8)?
   A) {(1,2), (2,3)}
   B) {(3,1), (2,4)}
   C) {(1,1), (3,2)}
   D) {(2,3), (1,2)}
   E) {(1,2), (2,5)}

Cognitive Analyzing

10. If a vector space has two linearly independent vectors, which of the following is true about the dimension of the space?
   A) The dimension is at least 2.
   B) The dimension is 2 only if the space is finite-dimensional.
   C) The dimension is exactly 1.
   D) The dimension is infinite.
   E) The space is 1-dimensional.

KHWAJA YUNUS ALI UNIVERSITY

School of Science & Engineering
Department of Computer Science and Engineering
Spring 2024 Final Examination
Program: B.Sc. in CSE (Batch:15th) 2nd Year 1st Semester
Course Title: Linear Algebra and Matrix Analysis
Course Code: CSE 0541-2101

Time: 10 Minutes **PART-A: MCQ** **Total Marks: 10**

Student's ID:

Date: **Invigilator's Signature:**

(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

CLO2	apply theory and calculation techniques for systems of linear equations (homogeneous/non homogeneous) to determine solvability and provide complete solutions and their structures.	02

Cognitive Remembering

1. Eigen values of a real matrix are always _____number.

   A) real
   B) positive
   C) real and imaginary
   D) negative
   E) imaginary

Cognitive Analyzing

2. Which is correct eigen vectors for the matrix $\begin{bmatrix} 0 & 0 & a \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$?

   A) $(0, a, 0)$
   B) $(a, a, 0)$
   C) $(0, a, a)$
   D) $(a, 0, 0)$
   E) $(0, 0, a)$

CLO 3	Perform on vector spaces, rank, nullity, Consistent, inconsistent, eigenvalues and eigenvectors and linear transformations	06

Cognitive Knowledge

3. A set of vectors in a vector space is called____.

   A) coordinate
   B) basis
   C) space
   D) unit
   E) dimension

Cognitive Evaluating

4. What is the value of the dimension of subspace $M_{2\times2}$,spanned by $\begin{pmatrix} 1 & -5 \\ -4 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ -1 & 5 \end{pmatrix}$ and $\begin{pmatrix} 2 & -4 \\ -5 & 7 \end{pmatrix}$?

   A) 1
   B) 2
   C) 3
   D) 4
   E) 6

Cognitive Knowledge

5. Which of the following solution is required for a system of equation is said to be inconsistent?

A) One
B) Two
C) Finite
D) No
E) Infinite

Cognitive Evaluating

6. What is the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$?

A) 0
B) 1
C) 2
D) 3
E) 4

Cognitive Evaluating

7. The system of linear equations $4x + 2y = 7$, $2x + y = 6$ has ________ solutions.

A) unique
B) exactly two distinct
C) exactly three distinct
D) no
E) infinite

Cognitive Knowledge

8. In linear equations, the finite and infinite set are classified as it's_______ set.

A) dimension
B) constant
C) variable
D) universal
E) solution

CLO4	apply the reduced (row) Echelon form of a matrix for determine linear combination, linearly dependent and independent, basis and dimension.	02

Cognitive Knowledge

9. In a matrix if not a single row can write as a linear combination of remaining rows, then what term is true for these matrix?

A) Identity dependence
B) Identity independence
C) linearly independent
D) linearly dependent
E) Cannot be determined

Cognitive Evaluating

10. How many linearly independent eigen vectors we get from the matrix $\begin{pmatrix} 2 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$?

A) -2
B) -1
C) 1
D) 2
E) 3

Signature of Examiner